

Number of outputs required

In many cases, a single output will be sufficient; however, multi-output supplies can sometimes deliver several important advantages.

Triple-output power supplies typically contain two higher-voltage outputs for analogue circuits (to power multi-voltage circuits or to create bipolar power supplies for testing bipolar analogue circuits) and a third output intended to power a digital circuit. For greatest flexibility, make sure that all three outputs are programmable.

If the DUT requires individual isolated power supply sections, a decision must be made to either configure multiple isolated supplies or buy a multi-output supply. The catch is that multi-output supplies can either have isolated outputs or output channels tied to a common point on their low side. When outputs are connected to the same common point, these are not suitable to power circuits that are isolated from each other.

For applications that require powering circuits up and down in a specific sequence, a multi-output supply with independently controllable outputs is usually better than a set of individual supplies.

For applications that require sourcing of more voltage or current beyond a single output's capability, some multi-output power supplies allow outputs to be combined in series or in parallel. Non-isolated multi-output power supplies cannot parallel the two outputs. To ensure flexibility, look for a multi-output power supply with isolated outputs.

Interfaces needed

There have been numerous types of

BENCHTOP POWER SUPPLIES

Make/brand	Product	Description
AIM-TTI Instruments 	Power supply, 3CH, 35V, 5A, adjustable (QL355TP)	2x (0 to 35V/0 to 3A or 0 to 15V/0 to 5A) plus 2.7/3.3/5.0 @ 1A
ARKSEN 	Precision variable adjustable dual-mode (A-003-PT-30145)	0-30V, 0-3A DC power supply
B&K Precision 	Benchtop power supplies 0-30V/0-3A 0-5V/0-3A, programmable triple (9130)	Channel1 0V to 30V, 3A Channel2 0V to 30V, 3A Channel3 0V to 5V, 3A
B&K Precision 	Benchtop power supplies triple-output DC power supply 60V/3A (9132B)	Channel1 0V to 60V, 3A Channel2 0V to 60V, 3A Channel3 0V to 5V, 3A
Keithley Instruments Inc. 	Benchtop triple-channel DC programmable power supply w/GPIB (2230G-30-1)	Channel1 0V to 30V, 1.5A Channel2 0V to 30V, 1.5A Channel3 0V to 6V, 5A
MASTECH 	Variable triple-output, 0-30V, 5A, LED display (HY3005F-3)	Dual adjustable outputs: 0-30V and 0-5A Fixed output: 5V and 3A

computer interfaces over the years with instrumentation. Two of the most popular have been IEEE-488, also known as general-purpose interface bus (GPIB), and RS232 serial communications. Network interfaces (for example, Ethernet) and USB interfaces have also been used.

One question to consider is: Would a digital I/O interface make it easier to generate fault status outputs or control an external relay or status lamp?

Placement of output connectors

Front-panel connections simplify frequent access and are more common with bench power supplies. Rear-panel connectors are generally considered superior for rack based auto-

mated test systems—these rarely require changes after setup. However, more manufacturers now offer power supplies equipped with both front- and rear-panel connections. This simplifies transition from bench experimentation to high-speed automated testing, because the same power supply suits both environments.

DIY bench power supply

A good power supply is very useful when working with electronics, but it can be costly. Any ATX (Advanced Technology eXtended) power supply can be plugged into the adaptor and can output 3.3V, 5V, 12V and -12V. (ATX is a motherboard developed by Intel in 1995 to improve on previous *de facto* standards.) It can be converted into a variable bench power supply. You can add female headers and banana jack (for appliances that draw more current) as output, or even a USB port.

For powering some DC motors, you can add variable output with an LM317, controlled by a potmeter. Although an ATX power supply should have a short-circuit protection built in, you can add output fuses as a safety measure. Make a case with plexiglass as it is nice to see the cable management inside.

In the same way you can combine LTC3780, which is a powerful 130W step-up/step-down converter, with a 12V 5A power supply to create an adjustable lab bench power supply in the range of 0.8V-29.4V and 0.3A-5A approximately.

You may add any standard make microcontroller and LCD to make it look like a modern power supply.